

Theorem B: the exact admissibility threshold of the closed form

A polyvortex mini-repository theorem monograph

Isaev Iskhak Khamzatovich (ORCID 0009-0003-7299-0701)

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Author	Isaev Iskhak Khamzatovich (ORCID 0009-0003-7299-0701)
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Abstract

The closed form of Theorem 3.1 — $r_k(t) = \sqrt{C_{Ch}}(1 + \varepsilon \cos(\omega t + 2\pi k/N))$ with the laws $\omega = (2\pi/T)e^{(C_{Ch}/\pi)}$ and $\varepsilon = 1/(e^{(C_{Ch}/\pi)} - 1)$ — describes a rotating configuration with positive radii not for all values of the topological invariant. This monograph proves the exact boundary: the radii are strictly positive at all times for all vortices if and only if $C_{Ch} > \pi \ln 2 = 2.177586090303602\dots$; at the threshold the amplitude equals exactly $\varepsilon = 1$ and each radius touches zero once per period; below it the radii change sign twice per period while the complex trajectory stays continuous. The proof is a chain of exact real-algebra equivalences. Certification by stage W5: on a grid of 200 values of $C_{Ch} \in [0.5, 10]$ the equivalence has zero violations, the amplitude at the threshold equals one to machine precision (error 0.0), and the sampled minimum radius matches the analytic $\sqrt{C_{Ch}}(1-\varepsilon)$ to $1.9 \cdot 10^{-7}$. A registered conclusion for the parent framework: the pair $(C_{Ch}, T) = (1, 2\pi)$ lies below the threshold — the structural registers of Theorem 3.1 are blind to it, the polar ones are not; parameter studies of Layer G should quote C_{Ch} relative to $C^{(*)}_{Ch} = \pi \ln 2$.

At a glance

Model	the closed form of Theorem 3.1 at arbitrary N (Layer G)
Headline result	$\varepsilon < 1 \Leftrightarrow C_{Ch} > \pi \ln 2 = 2.177586090303602$
Verification	stage W5: 200 grid points, 0 violations; min radius 1.9e-7
Registered fact	the parent default $(C_{Ch}, T) = (1, 2\pi)$ is below the threshold
Formalization queue	TB1 — the cheapest candidate (Mathlib / Coq Reals)

1. Introduction and setting

The topological invariant C_{Ch} enters the closed form of Theorem 3.1 twice: as the radius scale $\sqrt{C_{Ch}}$ and — through the amplitude law $\varepsilon(C_{Ch}) = 1/(e^{(C_{Ch}/\pi)} - 1)$ — as the depth of the radial modulation. As C_{Ch} grows, the amplitude decreases monotonically from $+\infty$ to zero, so the question of for which C_{Ch} the polar picture of the solution remains physically meaningful reduces to one exact inequality. The polar registers — radius positivity, the regular-N-gon picture, the gauge combination with $A_{\theta} = 1/r$ — are sensitive to the sign of the radius; the structural registers (the $2\pi/N$ separation, periodicity, pinned literals) are not. This monograph delivers the boundary between these two worlds in closed form, with a numerical certificate attached.

2. Statement and proof

Theorem B (admissibility). *The closed form of Theorem 3.1 (and of Hypothesis H1 at any N) has strictly positive radii for all times and all vortices if and only if*

$$C_{Ch} > C_{Ch}^* := \pi \ln 2 \approx 2.177586090303602.$$

At $C_{Ch} = C_{Ch}^*$ the amplitude equals exactly $\varepsilon = 1$ and each radius touches zero once per period; below the threshold each radius changes sign twice per period — on the complex plane the trajectory is continuous ($z_k = r_k e^{i\theta_k}$) is insensitive to the phase flip (π), but the polar registers degenerate.*

Proof. The minimum of $1 + \varepsilon \cos \varphi$ over a period equals $1 - \varepsilon$, so the positivity of all radii at all times is equivalent to $\varepsilon < 1$. The chain of exact equivalences follows:

$$\varepsilon < 1 \Leftrightarrow \frac{1}{e^{C_{Ch}/\pi} - 1} < 1 \Leftrightarrow e^{C_{Ch}/\pi} > 2 \Leftrightarrow C_{Ch} > \pi \ln 2,$$

using the strict monotonicity of $\exp$ and the mutual monotonicity of $\exp$ and $\ln$ on the positive half-axis. At $\varepsilon = 1$ the minimum of $1 + \cos \varphi$ is attained at the single point $\varphi = \pi$ — the radius touches zero once per period; at $\varepsilon > 1$ the cosine crosses the level $-1/\varepsilon$ twice per period — the radius changes sign twice. $\square$

3. Numerical certification (stage W5)

Stage W5 verifies the theorem on a uniform grid of 200 values $C_{Ch} \in [0.5, 10]$ through three independent registers. First, the pointwise equivalence $(C_{Ch} > C_{Ch}^*) \Leftrightarrow (\varepsilon < 1)$ — zero violations over the whole grid. Second, the amplitude at the threshold: $\varepsilon(\pi \ln 2) = 1$ with error 0.0 — a consequence of the exact identity $e^{\ln 2} = 2$. Third, the minimum radius: for all admissible grid points the sampled minimum of $r_k(t)$ over a 4096-node grid matches the analytic $\sqrt{C_{Ch}(1 - \varepsilon)}$ with the maximum absolute error $1.9 \cdot 10^{-7}$ — the uniform-grid resolution bound (quadratic in $1/4096$), not a model error.

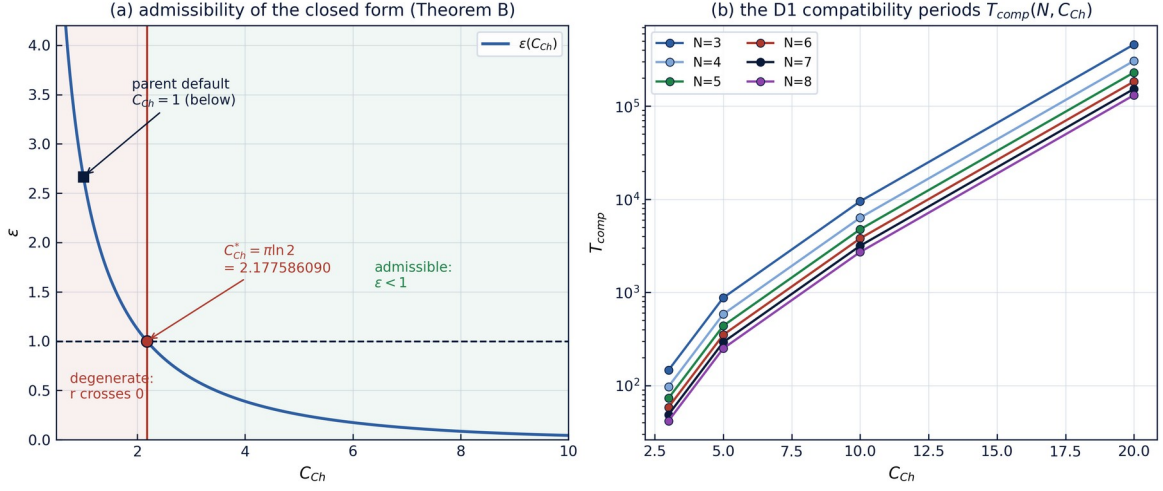


Fig. 1 — Left: the amplitude law $\varepsilon(C_{Ch})$ with the exact boundary $\pi \ln 2$, the degenerate and admissible zones and the parent default $C_{Ch} = 1$. Right: the D1 compatibility periods (Lemma D).

4. A registered fact about the parent default

The registered pair of the parent ladder, $(C_{Ch}, T) = (1, 2\pi)$, lies below the threshold: $\varepsilon(1) \approx 2.668 > 1$. The conclusion is careful but important: in this zone the closed form remains a perfectly well-defined complex choreography — the $2\pi/N$ separation, the periodicity and the literals $\omega = e^{i(1/\pi)}$ hold bit-for-bit (stage W1) — but the polar picture "an N -gon of radius $\sqrt{C_{Ch}(1 + \varepsilon \cos)}$ " is degenerate, the radii sweep through zero. The Section-6 diagnostic $C_{Ch}(t) = r^2(\dot{\theta} - q/r)$ degenerates at the crossing instants, which is why in the parent ladder it is report-only rather than pass/fail — this mini-repository reproduces the same honesty convention (diagnostic drift 15.01 over $[0, 100 T]$ at the default).

The practical reading for Layer-G parameter studies: the non-degenerate regime opens at $C_{Ch} > \pi \ln 2$, and C_{Ch} should be quoted relative to C_{Ch}^* . This makes Theorem B an exact, decidable delivery to the parent roadmap item T5, "the admissible region of the closed form".

5. Relation to the formalization roadmap

In the formalization queue Theorem B appears as item **TB1** and feeds the parent item **T5**. It is the cheapest formalization candidate: the statement is one inequality, the proof is the monotonicity of $\exp$ and $\ln$, available both in Mathlib and in Coq Reals; the boundary constant $\pi \ln 2$ admits a decidable rational certificate (e.g. the interval $[2.1775, 2.1776] \ni \pi \ln 2$ with rational endpoints). The numeric protocol W5 is the reference oracle of a future port; the grid equivalence is already verified by a test of the suite.

References

1. TRIVORTEX, verification/README.md — the V1–V4 ladder and the Theorem 3.1 registers.
2. H. Aref, P. K. Newton, M. A. Stremler, T. Tokieda, D. L. Vainchtein, Vortex crystals, *Advances in Applied Mechanics* 39 (2003) 1–79.
3. Mini-repository monograph: polyvortex/docs/monograph/monograph_EN.md, Section 5.